WiSH Analog HW — Pre-Work Instructions for Mentees

Overview

Welcome to the **WiSH** Analog Program!

This distinguished program, adapted from the training curriculum for new college graduates at Texas Instruments India, focuses on linear integrated circuits and MOS characteristics. Its purpose is to deepen your intuition and strengthen your fundamentals in circuit design which is the key building blocks for a successful engineering career.

Pre-Program Preparation
We will provide video recordings of the course delivered by TI Analog expert Anand Udupa. Please view these videos in advance of the live sessions. Doing so will enable you to grasp the concepts more quickly when the Subject Matter Experts (SMEs) present the material.

Daily Schedule
Each day consists of three structured sessions:

1. **Lecture Session** – SMEs present a curated set of slides, covering theory. They will solve several example problems, reinforcing analog fundamentals. At the end, you will receive ten practice problems to complete before the next day.

2. **Simulation Session** – SMEs conduct live walkthroughs of circuit simulations. You will be assigned a couple of simulation tasks to complete, which will be discussed in the following day's session.

3. **Practice Test Solution Session** – SMEs review selected problems from the previous day's practice problems, provide solutions, and address any questions you may have.

How to Maximize Your Learning
- Watch the video lectures prior to each class.
- Actively participate in live sessions and ask questions.
- Complete all simulation exercises and practice problems before the next day's class.
- Complete additional practice tests

Consistent, rigorous engagement will not only enhance your mastery of analog design but also position you for strong performance on the final test—an assessment that carries significant weight in the internship selection process.

Schedule Reference

Date	Day	Primary Topics
6/1	Monday	Voltage & Current Sources & RC Circuits Introduction to QUCS Simulator tool
6/2	Tuesday	Inductor Circuits Simulation Exercises Solution session for Voltage & Current Sources & RC Circuits
6/3	Wednesday	Fourier & Laplace Domain Analysis – Part 1 Simulation Exercises Solution session for Inductor Circuits
6/4	Thursday	Fourier & Laplace Domain Analysis – Part 2 Simulation Exercises Solution session for Fourier & Laplace Domain Analysis
6/5	Friday	Diode Circuits Simulation Exercises Solution session for Fourier & Laplace Domain Analysis
6/8	Monday	Buffer Circuits Simulation Exercises Solution session for Diode Circuits Additional Practice Test Solution Videos
6/9	Tuesday	Operational Amplifiers – Part 1 Simulation Exercises Solution session for Buffer Circuits
6/10	Wednesday	Operational Amplifiers – Part 2 Simulation Exercises Solution session for Op-Amp Additional Practice Test Solution Videos
6/11	Thursday	MOS Characteristics Simulation Exercises Live problem-solving sessions on MOS
6/12	Friday	Proctored WiSH Final Test

🗓️ Week2 — Pre-Work

📖 What to Expect

We begin the week with the fundamentals of voltage and current sources and RC circuits. On Day 1, the lecture will briefly cover both topics. Therefore, we highly recommend reviewing the corresponding video material beforehand. Following the below course schedule, plan to watch each topic's video prior to the day it is slated for instruction.

📺 Video Content

- Voltage & Current Sources - https://youtu.be/LOV9mxILVRc
- RC circuits - https://youtu.be/U71Do4SeG_4
- Laplace & Fourier Transforms - https://youtu.be/vZeT6lnSDww
- Inductors - https://youtu.be/LVy0EBMh0Ds
- Diodes - https://youtu.be/uvHJppwngdo
- Buffers - https://youtu.be/nrKk3mRAk90
- OpAmp - https://youtu.be/1dJIWzOfxzw
- MOS - https://youtu.be/r7-O_7Mlo1k

🗓️ Week2 — QUCS Simulator Setup Guide

Downloading this simulator is mandatory before the day 1 of the class. On day 1, how to use the QUCS simulator will be taught, so having this in your computer is a MUST.

- Download from https://qucs.sourceforge.net/ .

  - After download, you can extract the contents by clicking on the ZIP file. We recommend that you extract the contents in a folder such as C:\Qucs.
  - After extracting the files, use the File Explorer to locate C:\Qucs\qucs-0.0.19-win32mingw482-asco-freehdl-adms. You will find the application qucs.bat. Double click on it to start the simulator. It is better to start qucs from qucs.bat than qucs.exe.
  - The same site i.e. 'qucs.sourceforge.net' can be used for installation on MAC laptops as well. Search "qucs download for MAC" to get the respective link.
  - NOTE: A few setups might be disabled for MAC users.

Alternative for the same: WINE tool. This should be used only if QUCS has some setup disable.

📺 Video Content

Below link shows how to use QUCS to make circuits and simulate.

WiSH QUCS Simulation

📅 Week2 — Additional Practice Problem Sets

- Voltage & Current Sources https://www.surveymonkey.com/r/CWB5GDK
- RC Circuits https://www.surveymonkey.com/r/HYQYV8P
- Inductor circuits https://www.surveymonkey.com/r/HY8HDMF
- Fourier Laplace https://www.surveymonkey.com/r/HY9RLH3
- Diode circuits https://www.surveymonkey.com/r/CW2WGWY

📅 Week3 —Additional Practice Problem Sets

- Buffers https://www.surveymonkey.com/r/CQZTSY6
- Op Amp https://www.surveymonkey.com/r/CQHHWQD

Solutions to the above additional practice problem sets will be shared on 5th June and 10th June respectively.

📅 Week4 (6/17) — Analog Lab Session

📖 What is the Analog Lab Session?

On this day, in the morning session, a lecture on Operational Amplifier and its applications will be presented. Before lunch, the problem statement for the analog lab will be shared. Having a good background on the Operational Amplifier will help. Few resources to help you prepare are given below.

📺 **Video Content**

[Precision labs series: Op amps | TI.com]

Check out the videos of

- Input Offset voltage and Input bias
- Common mode rejection and Power supply rejection

📖 Other Resources

[TL082 data sheet, product information and support | TI.com]

[Hantek Oscilloscope]

[Hantek Oscilloscope Manual]

💡 **Tip:** If you run into any installation issues, reach out to your assigned mentor before the classroom session. You/assigned mentor can reach out to Ramanujam Thodur (tmrj@ti.com) for any further help.